

A Simulation Platform for Comparing Visual Guidance Strategies in Robotic Surgery Training

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I. INTRODUCTION

Robot-assisted telesurgery requires operators to control manipulators with novel dynamics, and overcome perceptual difficulties associated with reduced depth perception. Training novices to adopt experts' target-focused gaze strategies can improve complex motor skill acquisition [1], with varying types of visual guidance approaches being implemented [1]–[3]. To support our long-term goal of studying gaze training and guidance to improve telesurgical skills, we designed a simulator with integrated eye tracking and state-driven visual guidance. Its interchangeable visualizations support comparisons under consistent task, control, and display conditions, making it a flexible platform for telesurgical gaze-coaching research.

II. SYSTEM AND VISUAL GUIDANCE

The platform is built on SurRoL [4] and uses two Touch devices (3D Systems, California, USA) for bimanual teleoperation of the left and right PSMs (Fig. 1). A custom Robot Operating System 2 (ROS 2) interface publishes both Touch poses at 300 Hz. The devices are mounted symmetrically with opposing 45° yaw and 30° roll rotations, providing sufficient workspace for near-vertical block grasping while avoiding the devices' mechanical joint limits.

The peg transfer task requires transferring a block with the left PSM to an intermediate position and then with the right PSM to the final target (Fig. 1, inset). SurRoL and teleoperation run on an Ubuntu workstation, while visual guidance is rendered on a Windows laptop and streamed in side-by-side stereo through Virtual Desktop to a Varjo Aero HMD. Binocular gaze is acquired through the Varjo C++ API at 200 Hz.

The current gaze guidance framework uses task state to determine the current instruction and spatial target. During “acquisition”, the target is the peg *block*, while for the “first transfer”, it is the *middle position*, the “second transfer”, it is the *target position*. Three guidance methods are used: text, point-of-interest, and foveation. Text provides state-dependent instructions and red error warnings at a fixed screen location. Point-of-interest and foveation follow the projected target, using a red circular marker or a clearly rendered circular region of view, with blurred surroundings.

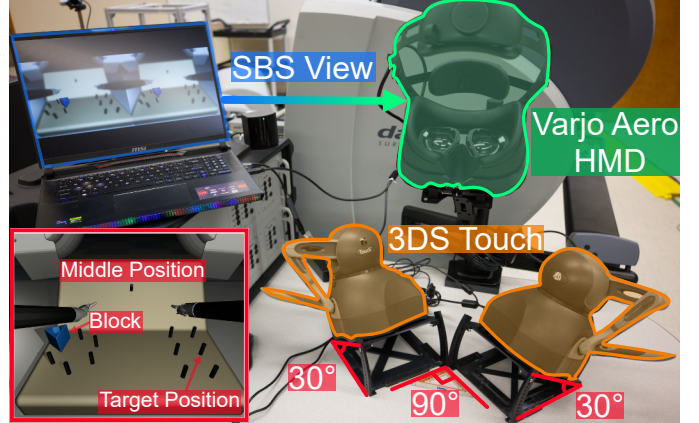


Fig. 1. System setup and simulated peg-transfer task. Two Touch devices provide bimanual teleoperation with annotated mounting angles. A Varjo Aero HMD supports display and eye tracking, and a laptop renders visual guidance. The inset shows the initial block, middle, and target positions. The SurRoL workstation is not shown. A system demonstration is available **Demo Video**.

III. FUTURE WORK

The current work establishes the telesurgery simulation platform but does not yet evaluate its effects on user attention or task performance. Future work will leverage Varjo's gaze and pupillometry recording capability to study the effect of different gaze guidance methods on gaze behavior, task performance, and subjective workload. We will also investigate how the operator's gaze can be used as a signal to modulate the training guidance, or task conditions to adapt difficulty.

REFERENCES

- [1] S. J. Vine, R. J. Chaytor, J. S. McGrath, R. S. Masters, and M. R. Wilson, “Gaze training improves the retention and transfer of laparoscopic technical skills in novices,” *Surgical Endoscopy*, vol. 27, no. 9, pp. 3205–3213, 2013.
- [2] D. R. Leff, D. R. James, F. Orihuela-Espina, K.-W. Kwok, L. W. Sun, G. Mylonas, T. Athanasiou, A. W. Darzi, and G.-Z. Yang, “The impact of expert visual guidance on trainee visual search strategy, visual attention and motor skills,” *Frontiers in human neuroscience*, vol. 9, p. 526, 2015.
- [3] Y. Feng, H. McGowan, A. Semsar, H. R. Zahiri, I. M. George, A. Park, A. Kleinsmith, and H. Mentis, “Virtual pointer for gaze guidance in laparoscopic surgery,” *Surgical endoscopy*, vol. 34, no. 8, pp. 3533–3539, 2020.
- [4] J. Xu, B. Li, B. Lu, Y.-H. Liu, Q. Dou, and P.-A. Heng, “SurroL: An open-source reinforcement learning centered and dvrc compatible platform for surgical robot learning,” in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2021.

This work was supported NIH Award Number 1R21EB036161-01.

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